1 This Week's Progress

1.1 General Items

- Researched methods to extract data from the Masimo RAD-97 device.
- Contacted Masimo support and coordinated with relevant personnel (Matt Gomez and Zach Calucchia) for licensing and trace software guidance.
- Reviewed available documentation, including manuals and online resources, to identify required adapters and cables.
- Tested connections and attempted preliminary data extraction setups.

1.2 Technical Progress

- Confirmed that RAD-97 data extraction is only possible with the Trace software.
- Identified part numbers for the software and required hardware:
  - Trace software: Part #9853
  - Null modem cable: Part #18313
  - Software provides a one-year license (price pending)
- Collected and annotated connector notes from the Masimo RAD-97 manual.
- Evaluated if previous RAD-8 data dumping methods can apply to RAD-97.
- Inserted two images showing relevant hardware setups and observations.
- Explored circuit testing with various chips, adjusting LED brightness without significant improvement.
- Began experiments with sample rate and pulse width to improve signal detection.

USB Port Baudrate	Sets the USB port communication speed.	921600	9600, 19200, 38400, 57600, 115200, 230400, or 921600

Figure 1: Required baud rate

Connector	Type
Ethernet	10/100 Mbps
Nurse Call	1/4 inch round female
USB	USB 2.0
NIBP Nib	Male Bayonet
NomoLine Capnography	Gas Inlet Connector

Figure 2: Basic connector notes

1.3 Challenges / Notes

- **Current issue:** Obtaining a stable and consistent output from the circuit is difficult and unable to extract data currently from Rad-97
- **Impact:** Progress on data extraction and signal validation is slowed; measurements are inconsistent.
- **Plan:** Continue experimenting with sample rate and pulse width adjustments, follow up with Masimo account manager for software access, and evaluate adapter compatibility for RAD-97.

2 Weekly Plans

- Receive confirmation and access from Masimo account manager for Trace software subscription.
- Acquire the required null modem cable and any additional serial adapters needed for RAD-97.
- Test RAD-97 data extraction using the adapted methods from RAD-8.
- Continue fine-tuning circuit testing parameters to achieve a stable signal output.
- Document findings, cable requirements, and configuration notes for reproducibility.